Inequalities — problem 7.12

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Let x_i, y_i, $1 \leq i \leq n$ be real numbers such that

$$x_1 \geq x_2 \geq \cdots \geq x_n \quad \text{and} \quad y_1 \geq y_2 \geq \cdots \geq y_n \qquad \text{(IMO 1975)}$$

Let $z_1, z_2, \ldots, z_n$ be any permutation of $y_1, y_2, \ldots, y_n$. Show that

$$\sum_{i=1}^{n}(x_i - y_i)^2 \leq \sum_{i=1}^{n}(x_i - z_i)^2.$$